

STATISTICAL TECHNIQUE_I

UNIT-IV

WHY STATISTICS?

- To Understand and Describe Data

Statistics helps summarize large amounts of information into meaningful forms.

- To Make Decisions Based on Evidence

Businesses, governments, doctors, and researchers use statistics to make informed decisions.

- To Identify Patterns and Relationships

Statistics helps discover trends and connections.

- To Predict the Future (Forecasting)

Using past data, statistics helps estimate future outcomes.

- To Evaluate Claims and Research

In science and social research, statistics provides proof.

- To Reduce Uncertainty

Life is full of uncertainty, but statistics helps measure and manage risk.

- To Support Fairness and Policy Making

Governments use statistical data for planning:

Without Statistics: Decisions are based on the intuitions and guess work

With Statistics: Decisions are based on facts and evidences

Real Life need of Statistics

Area of Life	Example	Purpose / Why Statistics is Needed
Health & Medicine	Testing a vaccine or medicine success rate	To check if treatment is effective and safe
Weather Forecasting	Predicting rain or temperature	To warn people and plan activities based on weather
Education	Calculating class average or grading	To evaluate performance and identify improvement areas
Business & Marketing	Customer surveys and product ratings	To understand customer needs and increase sales
Government & Planning	Census data (population, income, literacy)	To plan resources like schools, hospitals, and roads
Finance & Banking	Loan approval and insurance premium calculation	To measure risk and avoid financial loss
Sports	Batting average or player performance stats	To analyze strengths and improve strategies
Technology & Apps	YouTube/Netflix recommendations	To personalize recommendations and improve user experience

Statistical Thinking and Analysis

Statistical thinking is the process of understanding the world using data, logic, and probability instead of assumptions or opinions.

Key Features of Statistical Thinking

- Focuses on **data** instead of guesswork
- Recognizes that **variation exists**
- Uses **probability** to understand uncertainty
- Helps draw meaningful **conclusions**

Statistical analysis is the actual process or method used to work with data and extract meaningful information.

It includes:

- Collecting data
- Organizing and summarizing data (using tables, graphs)
- Applying formulas or tests
- Interpreting results

Feature	Statistical Thinking	Statistical Analysis
Meaning	Way of reasoning using data and evidence	Process of applying methods to analyze data
Focus	Understanding patterns, variation, and uncertainty	Calculating, summarizing, and interpreting numerical results
Purpose	To make thoughtful, evidence-based decisions	To extract, measure, and confirm information from data
Example	Realizing that exam scores depend on study habits and many other factors	Computing averages, correlation, or creating charts to study scores

Types of Statistics

Descriptive Statistics

These methods are used to **summarize and describe** data.

- Mean, median, mode
- Range, variance, standard deviation
- Tables, bar graphs, pie charts

Inferential Statistics

These methods use data from a **sample** to make conclusions or predictions about a **larger population**.

- Hypothesis testing
- Confidence intervals
- Correlation and regression
- t-test, chi-square test, ANOVA

Importance, Scope and Limitations of Statistics

Aspect	Details	Examples / Key Points
Importance of Statistics	Helps in summarizing and analyzing data for better understanding and decision-making.	• Decision-making • Simplifies complex data • Forecasting and planning Scientific research • Policy formation
Scope of Statistics	Used across various fields and includes collection, classification, presentation, analysis, and interpretation of data.	• Economics (national income, inflation) • Business (market surveys, quality control) • Education (grading, research) • Medicine (clinical studies) • Government (census, planning)
Limitations of Statistics	Statistics has constraints and cannot fully explain qualitative or emotional factors; accuracy depends on proper data.	• Cannot measure emotions exactly • Based on averages that may hide variations • Results depend on reliable data • Cannot prove cause-effect relationships • Can be misinterpreted or misused

Collection of Data

Type	Meaning	Example
Primary Data	Data collected directly by the investigator for a specific purpose	Survey, interview, observation
Secondary Data	Data collected by someone else and used by the investigator	Census reports, textbooks, newspapers

Method	Description	Example
Personal Interview	Data collected by asking questions face-to-face	Interviewing farmers about crop production
Questionnaire / Survey	Written questions answered by respondents	Google form survey on student stress
Observation	Data collected by watching events or behavior	Counting vehicles at a traffic signal
Telephone Interview	Data collected over phone calls	Calling customers for feedback
Experimentation	Data collected through controlled experiments	Testing medicine on sample groups

Collection of Secondary Data

Source of Secondary Data	Type	Examples
Government Publications	Published	Census reports, economic surveys, ministry reports
Non-Government Organizations (NGOs)	Published	Research reports, survey findings, statistical reports
International Organizations	Published	WHO reports, IMF statistics, UNESCO and World Bank data
Books and Textbooks	Published	Academic books, reference books, research literature
Journals and Research Papers	Published	Medical journals, education journals, scientific research reports
Newspapers and Magazines	Published	Articles, financial updates, statistical reports
Company/Business Records	Unpublished	Employee records, sales data, internal reports
Private Documents	Unpublished	Letters, diaries, personal records, case files
Websites and Online Databases	Digital Source	Government websites, Google Scholar, e-libraries

Classification of Data

Type of Classification	Meaning	Example Category	Example Data
Qualitative (Attribute) Classification	Data grouped based on qualities or characteristics, not numbers	Gender, Religion, Education Level	Male/Female, Hindu/Christian, Graduate/Postgraduate
Quantitative Classification	Data grouped based on numerical values	Age, Income, Height	Age groups like 10–20, 21–30; Income ₹10,000–₹50,000
Chronological Classification	Data arranged in order of time	Year, Month, Day	Population in 2015, 2016, 2017...
Geographical Classification	Data arranged by place or location	State, Country, District	Rainfall in Odisha, Tamil Nadu, Kerala
Conditional / Manifold Classification	Classification based on two or more characteristics	Gender + Literacy, Age + Occupation	Male–Literate, Female–Illiterate

Method of Data Collection

Method	Definition	Class Interval Style	Suitable for	Example
Exclusive Method	When data are classified so that the upper limit of one class becomes the lower limit of the next class.	Upper limit not included in the class.	Continuous data (measurements like height, marks, age).	0–10, 10–20, 20–30, 30–40
Inclusive Method	When both lower and upper limits are included within the same class interval.	Upper limit included in that class.	Discrete data (whole numbers like number of students, accidents, families).	0–9, 10–19, 20–29

Types of Graph

Type of Graph	Description	Used For
Bar Chart	Uses bars to show quantities	Qualitative or discrete data
Histogram	Rectangles with no gaps	Continuous numerical data
Pie Chart	Circular chart divided into sectors	Percentage or proportional data
Line Graph	Shows values with points connected by lines	Time series data (trends)
Frequency Polygon	Line graph drawn by joining midpoints of class intervals	Continuous frequency distribution
Ogive (Cumulative Frequency Curve)	Line graph showing cumulative frequency	Finding median, quartiles, percentiles